

BANO MARIAM

✉ banomariam97@gmail.com

☎ 0322-4496970

📍 Lahore cantt, PCHS, Pakistan

🌐 banomariam

SUMMARY

Software Engineering student with a strong focus on AI/ML and Computer Vision, experienced in building intelligent systems using deep learning and OCR techniques. Skilled in data preprocessing, model development, and real-world AI applications through academic and internship projects. Passionate about applying machine learning to solve practical problems and continuously expanding technical expertise.

EDUCATION

COMSATS University Islamabad, Lahore Campus

Bachelor of Science in Software Engineering

2022 - 2026

WORK EXPERIENCE

CacheLogic

Software Engineer Intern

July-2025 - present

Axtra Studio

Software Engineer Intern

February-2025 - May-2025

PROJECTS

Car Inspect – AI-Powered Vehicle Inspection System (Industrial FYP-CacheLogic)

- Developed a smartphone-based computer vision system for automated vehicle inspection.
- Designed a physics-informed repaint detection module using FFT, orientation entropy, GLCM, and CLAHE to identify aftermarket paint resprays without specialized hardware.
- Built a VIN and chassis authentication pipeline using YOLOv8, OCR ensembles, and image enhancement techniques for accurate VIN extraction across multiple formats.
- Implemented VIN tampering detection using Error Level Analysis (ELA), texture analysis, edge detection, and anomaly detection to identify grinding, restamping, and chassis manipulation.
- Developed a structural damage assessment system using YOLOv8 ensembles for automated detection of dents, scratches, and body damage.
- Applied advanced image preprocessing techniques, including super-resolution, deskewing, contrast enhancement, reflection removal, and noise reduction, to improve model performance.

RAG Document Intelligence Assistant

- Built an end-to-end RAG system for PDF, DOCX, and Markdown ingestion with semantic search and conversational QA.
- Implemented text extraction, overlap-aware chunking, all-MiniLM-L6-v2 embeddings, and ChromaDB vector similarity search.
- Developed FastAPI backend with document ingestion, querying, management, and source-grounded retrieval APIs.
- Integrated the Anthropic Claude API for context-aware, cited responses with retrieval-constrained generation to reduce hallucinations.
- Designed modular architecture for maintainable and extensible AI pipelines.

CERTIFICATE

AI Practitioner: Ready to Use and Shape AI – IBM [🔗](#)

Supervised Machine Learning: Regression and Classification [🔗](#)

Python Course-Kaggle [🔗](#)

SKILLS

- **Programming Languages:** Python, Java, C++, Dart
- **AI/ML:** Deep Learning, Computer Vision, LLMs, RAG, OCR, Semantic Search
- **Libraries:** NumPy, Pandas, OpenCV, Scikit-learn, TensorFlow, YOLOv8, Sentence Transformers, ChromaDB
- **Framework & Tools:** Django, FastAPI, Flutter, Firebase, REST APIs, GitHub, Docker, Google Colab, VS Code
- **Core Concepts:** Data Preprocessing, Feature Extraction, Model Training, Image Processing